

Microservices

<https://github.com/f1rset/Microservices>

Inside repository just run:

`docker compose up (-d for silent)`

1. Services Running

```

Microservices
1: docker compose up --build
   h.i.p.i.PartitionStateManager): [172.20.0.4]:5701 [dev] [5.3.8] Initializing cluster partition table arrangement.
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-1:50051
logging-1-1      | INFO:logging-service: [gRPC] Received and stored: 794b5822-86e8-41cb-9f98-42588cb676ba -> msg
1
facade-service-1 | INFO: 172.20.0.1:41580 - "POST /proxy?msg=msg1 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-1:50051
logging-1-1      | INFO:logging-service: [gRPC] Received and stored: 1dae89d7-477a-42c8-9e4e-c69657ea5ce6 -> msg
2
facade-service-1 | INFO: 172.20.0.1:41582 - "POST /proxy?msg=msg2 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-2:50051
logging-2-1      | INFO:logging-service: [gRPC] Received and stored: fa3793c6-22f7-4568-9372-1aabb45a11f6 -> msg
3
facade-service-1 | INFO: 172.20.0.1:41588 - "POST /proxy?msg=msg3 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-2:50051
logging-2-1      | INFO:logging-service: [gRPC] Received and stored: 4d3c51fc-2c37-40ce-9dae-e487ff3507f4 -> msg
4
facade-service-1 | INFO: 172.20.0.1:41598 - "POST /proxy?msg=msg4 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-1:50051
logging-1-1      | INFO:logging-service: [gRPC] Received and stored: 26aac447-0c21-4a70-8041-ef9fc830932d -> msg
5
facade-service-1 | INFO: 172.20.0.1:41602 - "POST /proxy?msg=msg5 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-2:50051
logging-2-1      | INFO:logging-service: [gRPC] Received and stored: 985cf7b6-2101-4fd2-ba4d-7b94586a80a5 -> msg
6
facade-service-1 | INFO: 172.20.0.1:41610 - "POST /proxy?msg=msg6 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-1:50051
logging-1-1      | INFO:logging-service: [gRPC] Received and stored: 3db150e9-3cf9-4f77-8835-e95c31e06892 -> msg
7
facade-service-1 | INFO: 172.20.0.1:41614 - "POST /proxy?msg=msg7 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-2:50051
logging-2-1      | INFO:logging-service: [gRPC] Received and stored: 286f4d1c-67f3-4287-9f3c-8764da80c3a8 -> msg
8
facade-service-1 | INFO: 172.20.0.1:41622 - "POST /proxy?msg=msg8 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-1:50051
logging-1-1      | INFO:logging-service: [gRPC] Received and stored: 2beb8fb3-a3bd-491d-85de-3eb9f1bd20c -> msg
9
facade-service-1 | INFO: 172.20.0.1:41626 - "POST /proxy?msg=msg9 HTTP/1.1" 200 OK
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-3:50051
logging-3-1      | INFO:logging-service: [gRPC] Received and stored: b2dc1bd3-efd2-4f0f-af12-f03114c44569 -> msg
10
facade-service-1 | INFO: 172.20.0.1:41638 - "POST /proxy?msg=msg10 HTTP/1.1" 200 OK

```

2. Messages can be read

```

→ Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg5, msg7, msg3, msg2, msg4, msg1, msg6, msg8, msg9, msg10 : \"Total messages processed and saved in DB: 1\"%"
→ Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg5, msg7, msg3, msg2, msg4, msg1, msg6, msg8, msg9, msg10 : \"Total messages processed and saved in DB: 2\"%"
→ Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg5, msg7, msg3, msg2, msg4, msg1, msg6, msg8, msg9, msg10 : \"Total messages processed and saved in DB: 3\"%"
→ Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg5, msg7, msg3, msg2, msg4, msg1, msg6, msg8, msg9, msg10 : \"Total messages processed and saved in DB: 4\"%"
→ Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg5, msg7, msg3, msg2, msg4, msg1, msg6, msg8, msg9, msg10 : \"Total messages processed and saved in DB: 5\"%"
→ Microservices git:(micro_hazelcast) x

```

3. Kill one logging service

```

logging-1-1 exited with code 137
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-1:50051
facade-service-1 | WARNING:facade-service:Service logging-1:50051 is unavailable. Error: StatusCode.DEADLINE_EXCEEDED
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-3:50051
logging-3-1      | INFO:logging-service: [gRPC] Received and stored: 8cf888f2-82cc-435a-a52f-dc96417fe426 -> msg
_after_crash
facade-service-1 | INFO: 172.20.0.1:51398 - "POST /proxy?msg=msg_after_crash HTTP/1.1" 200 OK

→ Microservices git:(micro_hazelcast) x docker stop microservices-logging-1-1
microservices-logging-1-1
→ Microservices git:(micro_hazelcast) x curl -X POST "http://localhost:8000/proxy?msg=msg_after_crash"
{"uuid":"8cf888f2-82cc-435a-a52f-dc96417fe426","status":"Stored in Hazelcast"}%
→ Microservices git:(micro_hazelcast) x

```

No data impact

```

Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg5, msg_after_crash, msg7, msg3, msg2, msg4, msg1, msg6, msg8, msg9, msg10 : \"Total messages processed and saved in DB: 6\"%"
Microservices git:(micro_hazelcast) x

```

4. Kill two hazelcast nodes (huge data loss because of no time to move data)

```

Microservices git:(micro_hazelcast) x docker stop microservices-hz1-1 microservices-hz2-1
microservices-hz1-1
microservices-hz2-1
Microservices git:(micro_hazelcast) x curl http://localhost:8000/proxy
"msg1, msg_after_crash, msg4, msg3, msg2 : \"Total messages processed and saved in DB: 7\"%"
Microservices git:(micro_hazelcast) x

```

```

logging-3-1 | INFO:hazelcast.connection:Connection(id=23, live=False, remote_address=None) closed. Reason:
Connection timed out
facade-service-1 | INFO:facade-service:Attempting to contact logging-service at logging-3:50051
logging-3-1 | INFO:logging-service: [gRPC] Returning all logs. Count: 5
counter-service-1 | INFO: 172.20.0.10:46052 - "GET /message HTTP/1.1" 200 OK
facade-service-1 | INFO:httpx:HTTP Request: GET http://counter-service:8002/message "HTTP/1.1 200 OK"
facade-service-1 | INFO: 172.20.0.1:36974 - "GET /proxy HTTP/1.1" 200 OK

```